Angel Canelo

Date of Birth: May 29, 1991
Nationality: Spanish



CONTACT INFORMATION

129, Samsung-ro, Yeongtong-gu, Suwon-si, Gyeonggi-do, Republic of Korea, R5 building 14th floor B-12

e-mail: angelcanelophd@gmail.com

Scholar: https://scholar.google.com/citations?user=wnc8LRUAAAAJ&hl=es

WORK

Samsung Electronics, Suwon, Korea
Staff Engineer
Deep Learning-based solutions for image enhancement and restoration
September 2024-Present

EDUCATION

Hanyang University, Seoul, Korea
Doctoral Degree in Electronics Engineering, 4.25/4.50 CGPA
August 2024
Thesis: Brain to artificial agents: Insect brain-inspired algorithms for vision-based navigation

University of Extremadura, Badajoz, Spain
Master's Degree in Industrial Engineering,
September 2017
Thesis: Modelling of the human vestibular system and integration in a simulator to study orientation and balance control

University of Extremadura, Badajoz, Spain
Bachelor's Degree in Electronics and Automation Engineering,
July 2015
Thesis: Design of a control system for traffic lights in isolated intersections

PUBLICATIONS

Papers

S Lee, E Park, A Canelo, et al. NTIRE 2025 Challenge on Efficient Burst HDR and Restoration: Datasets, Methods, and Results. *Proceedings of the Computer Vision and Pattern Recognition Conference*, 1002-1017 (2025).

A. Canelo, Y. Kim, J. Park, and A. J. Kim. A visual efference copy-based navigation algorithm in Drosophila for complex visual environments. *eLife* **13** (2024).

A. Canelo, I. Tejado, J. Traver, J. Prieto-Arranz, C. Nuevo-Gallardo, and B. Vinagre. A Human-Inspired Simulator for the Study of Orientation and Balance Control. *WSEAS Transactions on Biology and Biomedicine* **16**, 10–20 (2019).

A. Canelo, I. Tejado, J.E. Traver, B. Vinagre, and C. Nuevo-Gallardo. Modeling of the Human Vestibular System and Integration in a Simulator for the Study of Orientation and Balance Control. *XXXIX Jornadas de Automática, Área de Ingeniería de Sistemas y Automática, Universidad de Extremadura*, 2018, pp 636–643.

Selected presentations

A. Canelo, A. J. Kim. A visual efference copy-based navigation algorithm in Drosophila for complex visual environments. Poster presented at: *The Asia Pacific Drosophila Neurobiology Conference 3;* 2024 Feb 27-Mar 01; Tokyo, Japan.

A. Canelo, J. Ha, and A. J. Kim. An insect vision-inspired artificial neural network for a drone flight control. Poster presented at: *The 18th Korea Robotics Society Annual Conference;* 2023 Feb 15-18; Pyeongchang, Korea.

A. Canelo, A. J. Kim. An insect vision-inspired flight control algorithm with an auto-tuned efference copy. Poster presented at: *Neuroscience 2022. Annual Meeting of the SOCIETY for NEUROSCIENCE;* 2022 Nov 12-16; San Diego, USA.

A. Canelo, A. J. Kim. An insect vision-inspired flight control algorithm with an auto-tuned efference copy. Poster presented at: *The 51st Annual Meeting of the KOREAN DROSOPHILA SOCIETY;* 2022 Jul 04-05; Gyeongju, Korea.

A. Canelo, A. J. Kim. Efference copy-based flight control models for multi-featured visual scenes. Poster presented at: *Dialogues on neural and machine intelligence. Champalimaud Research Symposium;* 2021 Oct 13-15; Lisbon, Portugal.

A. Canelo, A. J. Kim. An integrative model of Drosophila flight control using efference copy. Poster presented at: *Neuroscience 2020 From Brain to Mind. The 23rd Annual Meeting of the Korean Society for Brain and Neural Sciences;* 2020 Nov 16-17; Seoul, Korea.

SCHOLARSHIPS

2018 - 2024 – Global Korea Scholarship (GKS)
2015 - 2016 – Spanish Ministry of Education Scholarship (MEC)

AWARDS

The best presentation at *The 51st Annual Meeting of the KOREAN DROSOPHILA SOCIETY;* 2022 Jul 04-05; Gyeongju, Korea: *An insect vision-inspired flight control algorithm with an auto-tuned efference copy*

The best presentation at *The 23rd Annual Meeting of the Korean Society for Brain and Neural Sciences;* 2020 Nov 16-17; Seoul, Korea: *An integrative model of Drosophila flight control using efference copy*

The best paper at *Las XXXIX Jornadas de Automática, Badajoz, 5-7 de Septiembre de 2018* in the category of Modeling, Simulation, and Optimization: *Modeling of the Human Vestibular System and Integration in a Simulator for the Study of Orientation and Balance Control*

The best academic transcript in the Master's degree: *Master's Degree in Industrial Engineering*

COMPETENCIES

Language – Spanish: Native level
English: TOEFL iBT certificate
Korean: TOPIK level 4 certificate
Japanese: Basic level

Project management – International Project Management Association IPMA level D Project Management Certificate

SELECTED TEACHING

Graduation Project Mentor Hanyang University 2021, 2022
Mentored students in the Deep Learning-based project: *Drone with insect-inspired vision*
Mentored students in the Deep Reinforcement Learning-based project: *Free-flight algorithm using Reinforcement Learning*

Biomedical Digital Engineering TA Hanyang University 2020
Supervised and graded laboratory practices, and mentored final term projects

Biomedical Systems Modeling TA Hanyang University 2019, 2020
Graded mid and final term exams, and gave seminar lectures

SKILLS

General – Computer Vision, Computational Neuroscience, Deep Learning, Reinforcement Learning, Control theory, Robotics, Embedded systems programming, PLC, PCB design, Project Management

Programming – Python, Matlab, C/C++, assembly, and Basic

Software – TensorFlow, PyTorch, OpenCV, Linux, MATLAB, SolidWorks, AutoCAD

PERSONAL INTERESTS	Languages, Music, Golf, Soccer